

Subject Name	BTL1123/BCL1123 Internet of Things
Assignment Title	Proposal Report & Video
Semester	May-Aug 2026
Release Date	As in LMS
Due Dates	As in LMS
Total Marks	100%

Continuous Assessment: Project Proposal Assignment

IMPORTANT INFORMATION

You are required to submit your work within the bounds of the University Infringement of Assessment Regulations. Plagiarism, paraphrasing and downloading large amounts of information from external sources will not be tolerated and will be dealt with severely. However, you should make full use of any source material, which would normally be an occasional sentence and/or paragraph (duly referenced) followed by your own critical analysis/evaluation.

The work must be entirely your own. You will receive ZERO (0) mark for work that is not your own. The safety of your assessments is your responsibility. You must not permit another student access to your work. Where referencing is required, APA referencing system must be used.

Learning Outcome

Apply IoT tools and devices in designing IoT solutions for real-world problems. [CLO 2]

Assignment Overview

This is an **individual assessment** carrying a weightage of 100% of the assignment marks. As a student in the Degree IoT ODL program, you are required to propose an IoT implementation to improve a specific area of your current accommodation area (e.g., a bedroom, living room, garden, community hall, or car porch).

This assignment is a **proposal**; no physical development or coding is required at this stage. The execution of this proposal will be conducted in your Final Project.

Task Instructions

- **Case Study:** Select one or two specific areas in your current accommodation. Take clear pictures of the area and elaborate on its current limitations or problems.
- **Hardware Selection:** Explore wokwi.com for available sensors and actuators. You may also custom-make a sensor in Wokwi (requires self-exploration, you can begin by referencing this [Wokwi blog post](#)).
- **System Design:** Sketch diagrams and draw a flowchart to illustrate how IoT can be implemented to improve the environment. Use pictures of sensors and actuators for illustrations.
- **Prototype Interface:** Design a prototype of the application interface to control your IoT system. It is highly recommended to use draw.io for your diagrams and UI wireframes. For tutorials on draw.io, you may refer to:
 - [Tutorial about draw.io](#)
 - [How to draw a flowchart](#)
 - [How to draw UI mockups/wireframes](#)
- **IoT 4 Layers Architecture:** Explain your proposed idea based on the IoT 4 Layers architecture. Describe the components of each layer in detail.
- **Uniqueness:** Highlight the special and unique idea you have introduced, comparing it to standard IoT applications already available in the market.
- **Video Presentation:** Record a presentation video (MAXIMUM 10 minutes) explaining the details of your proposal and highlighting the main points. **Ensure your camera is turned on and your face is visible throughout video. Marks will be deducted if the camera is off during the presentation.**

Report Format

- Your proposal report must include the following sections:
 1. **Cover Page**
 2. **Executive Summary:** A brief summary of the proposal.
 3. **Target Environment:** Introduction to the area (include pictures/photos here) and highlight the limitations/problems.
 4. **Proposed Solution and Illustrations:** Hardware selections, system architecture, flowchart, and dashboard UI prototype.
 5. **IoT 4 Layers Explanation:** Detailed explanation of the 4 layers in the context of your project.
 6. **Summary**
 7. **References:** Include citations for any source of references used (APA format).
 8. **Link to access presentation video (shared file)**
- Make sure all information required is accurate. All information retrieved from the Internet or textbook must be cited accordingly.
 1. Reference from Wikipedia is not allowed.
 2. Create a clean, logically structured Table of Contents for the proposal, reflecting the sections in the task instructions.

FORMATTING INSTRUCTIONS

- Font
 - Use Times New Roman font, 12-point font style
 - Use bold, italics, and underline in an appropriate manner
- Paragraph
 - Alignment Justified
 - Indentation First line
 - Use 1.5 line spacing
- Styles
 - Use headings and subheadings.
- Page layout
 - Orientation portrait and landscape
 - Page break and section break
 - Different Header and footer for different sections
- Picture tools
 1. Format picture
 2. Insert caption
- Use APA referencing and citations.

Submission Instructions

- Upload the report (docx).
- Upload the video to an external platform (share drive) and include the link in your report.
- Ensure the video is accessible. It is the student's responsibility to make sure the marker can view it.
- Note: If the marker cannot open the video due to restricted access permissions, a grade of 0 marks will be given for the video presentation component.
- The work must be entirely your own. Plagiarism will result in ZERO (0) marks.
- Submission Deadline and Late Policy:
 - Please submit your assignment by or before the deadline.
 - **LATE SUBMISSION** will be **DEDUCTED 10% penalty A DAY** up to a **maximum of FIVE (5) days**, were subsequently, the assignment will be awarded **ZERO (0)** mark.
 - **ZERO (0)** mark will be given if there is any form of copying/plagiarism.
 - **ZERO (0)** mark will be given if you fail to submit your work or fail to submit all the necessary files.

Assignment Marking Scheme

Assignment Question Summary

Students are required to individually propose an IoT implementation to improve their current accommodation (case study). They must identify limitations with photos, select hardware (via Wokwi), sketch system diagrams and flowcharts, design a UI prototype (via draw.io), and explain the system using the IoT 4-layer architecture. A 10-minute presentation video is also required.

Marking Scheme

Your work will be assessed based on the following criteria.

- **Problem Identification (Environment) (15%):** Clear identification of the problem with photographic evidence and solid justification.
- **Proposed Solution & Hardware (20%):** Viable solution with appropriate sensor/actuator selection from Wokwi.
- **System Design (Diagrams & Flowchart) (20%):** Clear, professional flowcharts and architecture diagrams illustrating the system.
- **UI Prototype (10%):** Intuitive and responsive UI wireframe visualizing user interactions.
- **IoT 4 Layers Architecture (15%):** Accurate explanation of the Edge, Connectivity, Cloud, and Application layers for your project.
- **Report Quality & References (10%):** Professional formatting, clarity, and correct APA citations.
- **Video Presentation (10%):** Clear, engaging, and persuasive explanation of your proposal within 10 minutes.